

„ZPR PWr – Zintegrowany Program Rozwoju Politechniki Wrocławskiej”

Virtual Reality, VR support in Unity

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wideo z wykładu:

https://pwr-edu.zoom.us/rec/share/kxClOTxSojwun0_2veLnWDU3qCrpWlwhW1O8IHm1t5LI3_nVhurjFsYZArWyw8Be.rb0_snWVaLgzTCiB

Lecture

1. **Silniki gier. Wprowadzenie do Unity. Pierwsza gra 2D.**
2. **Mechanika 2D, animacje.**
3. **Oświetlenie, tekstury, materiały. Pierwsza gra 3D.**
4. Narzędzia wspierające, np. Blender.
5. **Zapis/odczyt danych, komunikacja sieciowa.**
6. **Modelowanie terenu. Generowanie terenu.**
7. **Sztuczna inteligencja w grach.**
8. Virtual Reality, wsparcie VR w Unity.
9. **Optymalizacje w Unity.**
10. Historia i klasyfikacja gier.
11. Prototypowanie gier. GDD.
12. Projektowanie poziomów gier.
13. Projektowanie gier dla różnych platform.
14. Testowanie gier. Test.



ROOM-SCALE



STANDING ONLY



VR Gaming Guide



Sitting
Down



Standing



Room
Scale

Ile to kosztuje?

cardboard

stand-alone HMD

HMD + desktop

Cheap

Mid-range

Premium

Price

\$15 - \$50

\$300 - \$600

\$800 - \$1,000

Headsets available

Smaller brands, usually running off smartphones

Oculus Quest 2, HP Reverb G2

HTC Vive Pro 2, Valve Index

Meta Oculus Quest 2 - **1600 PLN**

HTC Vive Pro 2 goggle - **3800 PLN**, cały zestaw (bez PC) - **6700 PLN**

Sony PlayStation VR2 - **3000 PLN**, z konsolą - **7000 PLN**



[How much is a VR headset? | GamesRadar+](#)

[The best VR headset in 2023: all the latest platforms compared | GamesRadar+](#)



XR

<https://docs.unity3d.com/Manual/XR.html>

XR is an *umbrella term* that includes the following types of applications:

Virtual Reality (VR): The application simulates a completely different environment around the user.

Mixed Reality (MR): The application combines its own environment with the user's real-world environment and allows them to interact with each other.

Augmented Reality (AR): The application layers content over a digital view of the real world.

Unity supported platforms

<https://docs.unity3d.com/Manual/XR.html>

Unity supports these platforms for:

ARKit

ARCore

Microsoft HoloLens

Windows Mixed Reality

Magic Leap

Oculus

OpenXR

PlayStation VR

Unity **does not** support XR on **WebGL**

Unity supported platforms

<https://docs.unity3d.com/Manual/VROverview.html>

Oculus

If you're developing for **Oculus Quest**, you need to install and load the **Oculus XR Plug-in for the Android build target**.

If you're developing for **Oculus Rift and Rift S**, you need to install and load the **Oculus XR Plug-in for the Windows build target**.

Windows Mixed Reality

If you're developing for a **Windows MR VR headset**, you need to install and load the **Windows XR Plug-in for the Universal Windows Platform**, and use Microsoft's Mixed Reality Toolkit ([MRTK 2.3](#)) to add mixed reality input and interactions.

Third-party supported platforms

Google VR → Google's [VR SDK Quickstart guide](#) for the latest on Google Daydream and Cardboard

OpenVR → Valve has released their OpenVR Unity XR Plug-in → **SteamVR (HTC Vive, Valve Index)**

The screenshot displays the Unity 2020.2.1f1 Personal interface. The main view is a 3D scene titled "Interactions_Example" showing a virtual environment with various interactive objects like a teleporter, a longbow, and a circular drive. The "SteamVR Input" window is open, showing the "Advanced Settings" tab. It lists "Action Sets" (default, platformer, buggy, mixedreality) and "Actions" (In, Out, Haptic). The "Inspector" panel on the right shows the "SteamVR Input Live View" for the "Not Bound" state, listing various actions and their status (e.g., default, platformer, buggy, mixedreality, ExternalCamera, LeftHand, RightHand, LeftFoot, RightFoot, LeftShoulder, RightShoulder, Waist, Chest). The "Hierarchy" panel on the left shows the scene's structure, including "Player", "Environment", "Teleport", "Simple Interactable", "Interesting Interactables", "LinearDrive", "Longbow", "CircularDrive", "Throwing", "Hints", "Hover Button", "Skeleton", and "Remotes". The "Project" panel at the bottom shows the "Assets" folder structure, including "Core", "Hints", "Longbow", "Poses", "Samples", "SnapTurn", "Teleport", "Materials", "Models", "Prefabs", "Scripts", "Squishy", "Textures", and "Interactio...".

SteamVR Input - Advanced Settings

Action Sets

- default
- platformer
- buggy
- mixedreality

Actions

In

- InteractUI
- Teleport
- GrabPinch
- GrabGrip
- Pose
- SkeletonLeftHand
- SkeletonRightHand
- Squeeze
- HeadsetOnHead
- SnapTurnLeft
- SnapTurnRight

Out

Haptic

Save and generate **Open binding UI**

Inspector - SteamVR Input Live View

Not Bound **Inactive**

Any

- default **Inactive**
- platformer **Inactive**
- buggy **Inactive**
- mixedreality **Inactive**
- ExternalCamera **Inactive**

LeftHand

- default **Inactive**
- InteractUI **False**
- Teleport **False**
- GrabPinch **False**
- GrabGrip **False**
- Pose **(0.4826, 0.5669, -0.8979) : (0.0231, -0.9885, -0.1208)**
- Squeeze **0.0000**
- HeadsetOnHead **False**
- SnapTurnLeft **False**
- SnapTurnRight **False**
- Haptic **-193 seconds since last used**

platformer **Inactive**

buggy **Inactive**

mixedreality **Inactive**

ExternalCamera **Inactive**

RightHand

- default **Inactive**
- InteractUI **False**
- Teleport **False**
- GrabPinch **False**
- GrabGrip **False**
- Pose **(0.1810, 0.5757, -0.8078) : (-0.0901, 0.7121, 0.0771)**
- Squeeze **0.0000**
- HeadsetOnHead **False**
- SnapTurnLeft **False**
- SnapTurnRight **False**
- Haptic **-129 seconds since last used**

platformer **Inactive**

buggy **Inactive**

mixedreality **Inactive**

ExternalCamera **Inactive**

LeftFoot

RightFoot

LeftShoulder

RightShoulder

Waist

Chest

New project

Editor Version: **2021.3.24f1** LTS ↕

☰ All templates

★ New

▣ Core

☰ Sample

🎓 Learning

SRP
3D (HDRP)
Core
VR
Core
AR
Core
First Person
Core
Third Person
Core

VR

Quickstart your Virtual Reality (VR) applications with a sample scene, assets, and the recommended packages and...

[Read more](#)

PROJECT SETTINGS

Project name
My VR test

Cancel

Create project

VR vs non-VR



jedyny aspekt, gdzie wygrał non-VR.

2. How **enjoyable and engaging** were the sensations of **moving, turning, crouching, hiding**?



Dlaczego?



VR locomotion: treadmill vs skating

[12 VR Locomotion Solutions/Concepts You Probably Haven't Heard Of](#)

[EKTO ONE Product Reveal](#) (covered above)

[A completely new way to walk in VR: CyberShoes](#) (8:15)

- problems with shoes and full body tracking

<https://www.facebook.com/reel/314097304394223>

Oculus in Unity

<https://assetstore.unity.com/packages/tools/integration/oculus-integration-82022>

→ <https://developer.oculus.com/documentation/unity/unity-package-manager/>

study case:

“How to handle head collisions in VR”

tylko **ZAJAWKA**
- całość na
Gry Komputerowe
II stopień,
spec. PSI

findings from an experiment
for a Master Thesis

by student Bartłomiej Stanasiuk
and me as supervisor

[#6 Kopel_Marek, IEAAIE2020](#)

study case:

“Which gameplay aspects impact the immersion in virtual reality games?”

tylko **ZAJAWKA**
- całość na
Gry Komputerowe
II stopień,
spec. PSI

findings from an experiment
for a Master Thesis

by student Marta Rutkowska
and me as supervisor

[#163 Kopel_Marek, ICCCI2021](#)